The documentation and process conversion measures necessary to comply with this document shall be completed by 13 February 2014.

INCH-POUND

MIL-PRF-19500/538G
13 December 2013
SUPERSEDING
MIL-PRF-19500/538F
10 February 2011

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER,
TYPES 2N6676, 2N6678, 2N6676T1, 2N6678T1, 2N6676T3, 2N6678T3, 2N6691, AND 2N6693,
JAN, JANTX, JANTXV, JANS, JANSM, JANSD, JANSP, JANSL, JANSR, JANSF, JANSG, JANSH, JANHCA,
JANKCA, JANKCAM, JANKCAD, JANKCAP, JANKCAL, JANKCAR, JANKCAF, JANKCAG, AND JANKCAH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, power transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type. Provisions for radiation hardness assurance (RHA) to eight radiation levels is provided for JANS and JANKC product assurance levels. RHA level designators “M”, “D”, “P“, “L”, “R”, “F”, “G”, and “H” are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Physical dimensions. See figure 1 (TO-3) - 2N6676, 2N6678; figure 2 (TO-61) - 2N6691, 2N6693, figure 3 (TO-254AA) - 2N6676T1, 2N6678T1, figure 4 (TO-257AA) - 2N6676T3, 2N6678T3, and figure 5 (JANHC and JANKC).

1.3 Maximum ratings.

| Types | $P_T$ $T_A$ = +25°C | $P_T$ (1) $T_C$ = +25°C | $R_{\theta JC}$ (2) | $V_{CBO}$ and $V_{CEX}$ | $V_{CEO}$ | $T_J$ and $T_{STG}$ | $V_{EBO}$ | $I_B$ | $I_C$ |
|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | V dc | V dc | °C | V dc | A dc | A dc |
| 2N6676, 2N6676T1 | 6 | 175 | 1.0 | 450 | 300 | -65 to +200 | 8.0 | 5 | 15 |
| 2N6678, 2N6678T1 | 6 | 175 | 1.0 | 650 | 400 | -65 to +200 | 8.0 | 5 | 15 |
| 2N6676T3 | 4 | (3) 125 | 1.3 | 450 | 300 | -65 to +200 | 8.0 | 5 | 15 |
| 2N6678T3 | 4 | (3) 125 | 1.3 | 650 | 400 | -65 to +200 | 8.0 | 5 | 15 |
| 2N6691 | 6 | 175 | 1.0 | 450 | 300 | -65 to +200 | 8.0 | 5 | 15 |
| 2N6693 | 6 | 175 | 1.0 | 650 | 400 | -65 to +200 | 8.0 | 5 | 15 |

(1) See figures 6 and 7 for temperature-power derating curves.
(2) See figures 8 through 11, thermal impedance curves.
(3) For TO-257 devices with typical mounting and small footprint, conservatively rated at 125 W and 1.3°C/W only.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .



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1.4 Primary electrical characteristics at $T_C$ = +25°C.

| Limits | $h_{FE1}$<br>$V_{CE}$ = 3 V dc<br>$I_C$ = 1 A dc<br>(1) | $h_{FE2}$<br>$V_{CE}$ = 3 V dc<br>$I_C$ = 15 A dc<br>(1) | $V_{BE(sat)}$<br>$I_C$ = 15 A dc<br>$I_B$ = 3 A dc | $V_{CE(sat)}$<br>$I_C$ = 15 A dc<br>$I_B$ = 3 A dc |
|---|---|---|---|---|
| | | | V dc | V dc |
| Min | 15 | 8 | | |
| Max | 40 | 20 | 1.5 | 1.0 |

| Limits | $\|h_{fe}\|$ | $C_{obo}$ | Switching (2) | | | | |
|---|---|---|---|---|---|---|---|
| | $V_{CE}$ = 10 V dc<br>$I_C$ = 1 A dc<br>f = 5 MHz | $V_{CB}$ = 10 V dc<br>$I_E$ = 0<br>100 KHz < f < 1 MHz | $t_c$ | $t_d$ | $t_r$ | $t_s$ | $t_f$ |
| | | pF | µs | µs | µs | µs | µs |
| Min | 3 | 150 | | | | | |
| Max | 10 | 500 | 0.5 | 0.1 | 0.6 | 2.5 | 0.5 |

(1) Pulsed (see 4.5.1).
(2) See figure 12 (pulse response circuits).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

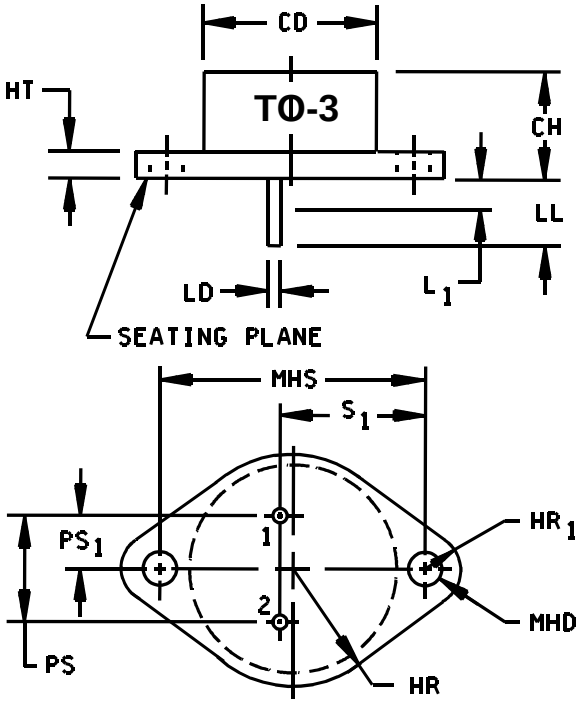
* (Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Ltr | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | | .875 | | 22.22 | 3 |
| CH | .270 | .380 | 6.86 | 9.65 | |
| HR | .495 | .525 | 12.57 | 13.34 | |
| $HR_1$ | .131 | .188 | 3.33 | 4.78 | |
| HT | .060 | .135 | 1.52 | 3.43 | |
| $L_1$ | | .050 | | 1.27 | 5, 9 |
| LD | .038 | .043 | 0.97 | 1.09 | 5, 9 |
| LL | .312 | .500 | 7.92 | 12.70 | 5 |
| MHD | .151 | .161 | 3.84 | 4.09 | 7 |
| MHS | 1.177 | 1.197 | 29.90 | 30.40 | |
| PS | .420 | .440 | 10.67 | 11.18 | 4 |
| $PS_1$ | .205 | .225 | 5.21 | 5.72 | 4, 5 |
| $S_1$ | .655 | 0.675 | 16.64 | 17.14 | 4 |

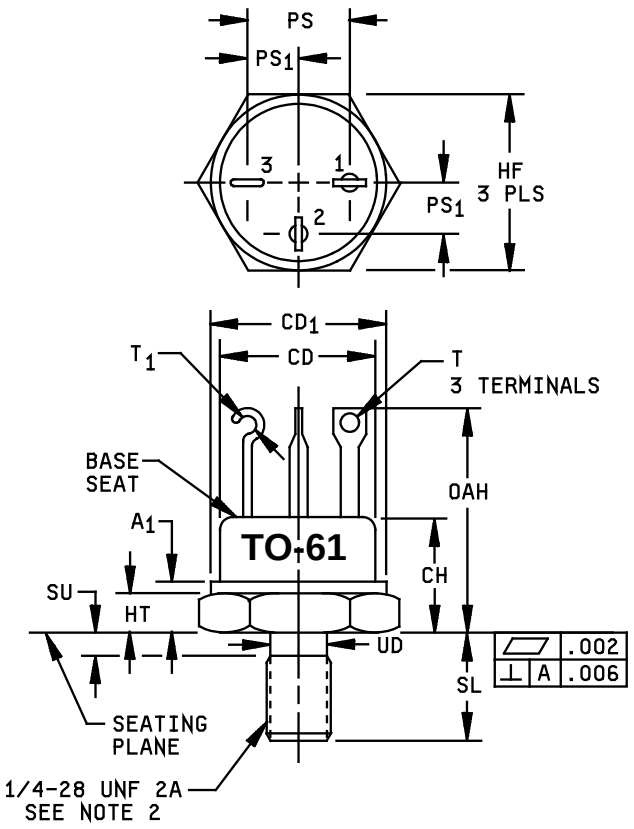



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Body contour is optional within zone defined by CD
4. These dimensions shall be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. Measurement shall be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. Terminal 1 is the emitter, terminal 2 is base. The collector shall be electrically connected to the case.
9. LD applies between $L_1$ and LL. Diameter is uncontrolled in $L_1$.
10. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (TO-3) for 2N6676 and 2N6678.



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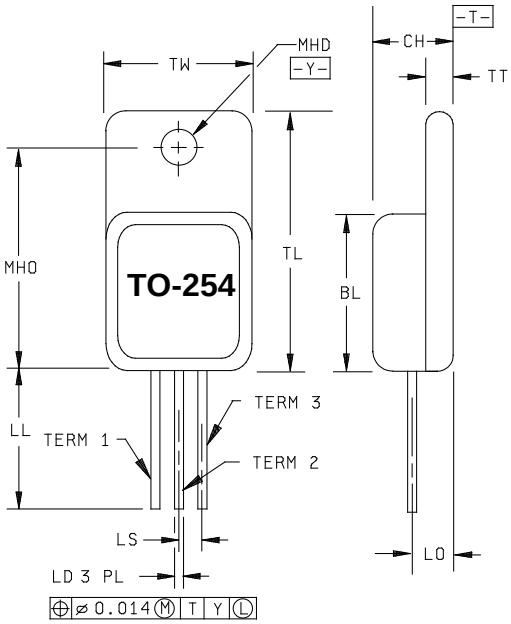

| Ltr | Dimension | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| $A_1$ | | .270 | | 6.86 | |
| CD | .570 | .610 | 14.48 | 15.49 | |
| $CD_1$ | .610 | .687 | 15.49 | 17.45 | |
| CH | .325 | .460 | 8.26 | 11.68 | |
| HF | .667 | .687 | 16.94 | 17.45 | |
| HT | .090 | .150 | 2.29 | 3.81 | |
| OAH | .640 | .875 | 16.26 | 22.22 | 4 |
| PS | .340 | .415 | 8.64 | 10.54 | 3, 6 |
| $PS_1$ | .170 | .213 | 4.32 | 5.41 | 3, 6 |
| SL | .422 | .455 | 10.72 | 11.56 | |
| SU | | .090 | | 2.29 | 7 |
| T | .047 | .072 | 1.19 | 1.83 | |
| $T_1$ | .046 | .077 | 1.17 | 1.96 | |
| UD | .220 | .249 | 5.59 | 6.32 | |

NOTES:
1. Dimensions are in inches, millimeters are given for general information only.
2. See NSB Handbook H28, “Screw-Thread Standards for Federal Services”.
3. The orientation of the terminals in relation to the hex flats is not controlled.
4. All three terminals.
5. The case temperature may be measured anywhere on the seating plane within .125 inch (3.18 mm) of the stud.
6. Terminal spacing measured at the base seat only.
7. This dimension applies to the location of the center line of the terminals.
8. Terminal - 1, emitter; terminal - 2, base; terminal - 3, collector. All leads are isolated from the case.
9. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions (TO-61) for 2N6691 and 2N6693.



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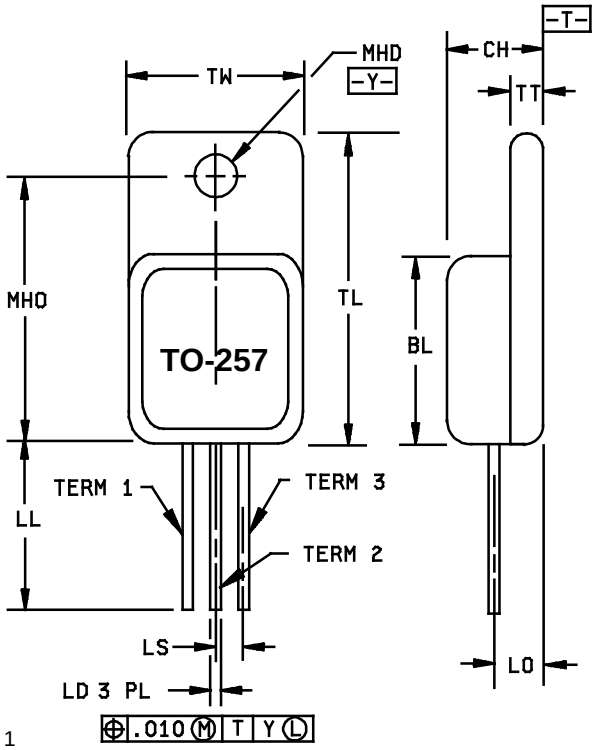

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .535 | .545 | 13.59 | 13.84 |
| CH | .249 | .260 | 6.32 | 6.60 |
| LD | .035 | .045 | 0.89 | 1.14 |
| LL | .510 | .570 | 12.95 | 14.48 |
| LO | .150 BSC | | 3.81 BSC | |
| LS | .150 BSC | | 3.81 BSC | |
| MHD | .139 | .149 | 3.53 | 3.78 |
| MHO | .665 | .685 | 16.89 | 17.40 |
| TL | .790 | .800 | 20.07 | 20.32 |
| TT | .040 | .050 | 1.02 | 1.27 |
| TW | .535 | .545 | 13.59 | 13.84 |
| Term 1 | Base | | | |
| Term 2 | Collector | | | |
| Term 3 | Emitter | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Methods used for electrical isolation of the terminals feedthroughs shall employ materials that contain a minimum of 90 percent AL2O3 (ceramic).
4. All terminals are isolated from case.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Dimensions and configuration for 2N6676T1 and 2N6678T1 (TO-254AA).



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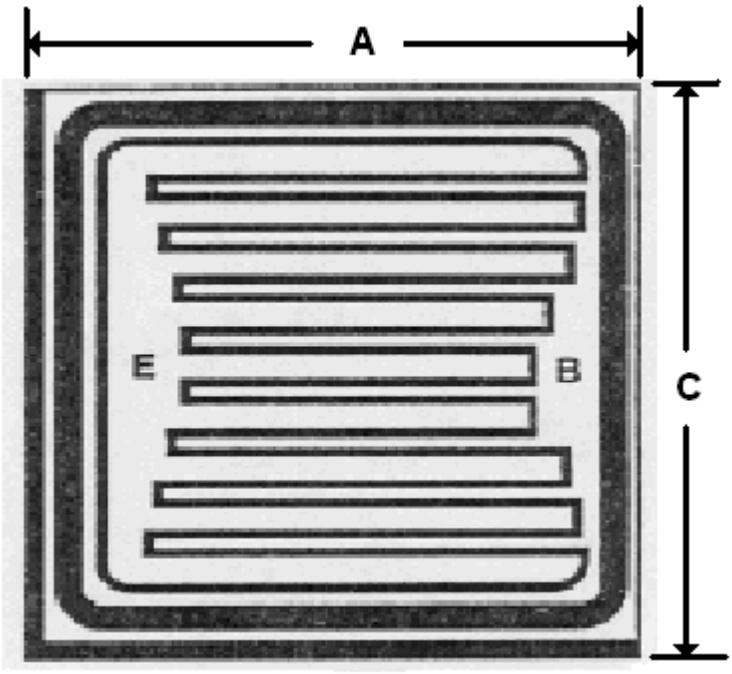
| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .410 | .430 | 10.41 | 10.92 |
| CH | .190 | .200 | 4.83 | 5.08 |
| LD | .025 | .035 | 0.64 | 0.89 |
| LL | .500 | .750 | 12.70 | 19.05 |
| LO | .120 BSC | | 3.05 BSC | |
| LS | .100 BSC | | 2.54 BSC | |
| MHD | .140 | .150 | 3.56 | 3.81 |
| MHO | .527 | .537 | 13.39 | 13.63 |
| TL | .645 | .665 | 16.38 | 16.89 |
| TT | .035 | .045 | 0.89 | 1.14 |
| TW | .410 | .420 | 10.41 | 10.67 |
| Term 1 | Base | | | |
| Term 2 | Collector | | | |
| Term 3 | Emitter | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Methods used for electrical isolation of the terminals feedthroughs shall employ materials that contain a minimum of 90 percent AL2O3 (ceramic).
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Dimensions and configuration for 2N6676T3 and 2N6678T3 (TO-257AA).



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| Ltr. | Dimensions | | | |
|---|---|---|---|---|
| | Inch | | Millimeters | |
| | Min | Max | Min | Max |
| A, C | .205 | .215 | 5.21 | 5.46 |

DESIGN DATA
Metalization:
Top: Aluminum 54,000 Å minimum, 60,000 Å nominal.
Back: Al/Ti/Ni/Au 10,000 Å minimum, 12,500 Å nominal.
Back side: Collector.
Chip thickness: .012 inch (0.305 mm) ±.002 inch (0.051 mm).
Bonding pad: B = .018 inch (0.46 mm) x .040 (1.02 mm).
E = .018 inch (0.46 mm) x .040 (1.02 mm).

* FIGURE 5. JANHCA and JANKCA (A version) die dimensions.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

$V_{CEX}$ - Collector cutoff voltage (dc) with specified circuit between base and emitter.
$R_{ISO}$ - Resistance between device case and leads

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 (TO-3) - 2N6676, 2N6678; figure 2 (TO-61) - 2N6691, 2N6693; figure 3 (TO-254AA) - 2N6678T1, 2N6676T1, and figure 4 (TO-257AA) - 2N6676T3 and 2N6678T3 herein, and figure 5.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Construction. These devices shall be constructed in a manner and using materials which enable the devices to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.

3.6 Marking. Marking shall be in accordance with MIL-PRF-19500. The radiation hardened designator M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I, II, and III).



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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening. Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) | Measurements | |
|---|---|---|
| | JANS | JANTX and JANTXV levels |
| (1) 3c | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.1) | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.1) |
| 9 | $I_{CEX1}$ and $h_{FE2}$ | $I_{CEX1}$ and $h_{FE2}$ |
| 11 | $I_{CEX1}$ and $h_{FE2}$<br>$\Delta I_{CEX1}$ = 100 percent of initial value or 500 nA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ±15 percent of initial value. | $I_{CEX1}$ and $h_{FE2}$<br>$\Delta I_{CEX1}$ = 100 percent of initial value or 500 nA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ±25 percent of initial value. |
| 12 | See 4.3.2 | See 4.3.2 |
| 13 | Subgroup 2 and 3 of table I herein;<br>$\Delta I_{CEX1}$ = 100 percent of initial value or 500 nA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ±25 percent of initial value. | Subgroup 2 of table I herein;<br>$\Delta I_{CEX1}$ = 100 percent of initial value or 500 nA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ±25 percent of initial value. |
| * 17 | For TO-254AA packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein. | For TO-254AA packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein. |

* (1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

* 4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, (and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. The thermal impedance limit shall comply with the thermal impedance graph on figures 8, 9, 10, and 11 (less than or equal to the curve value at the same $t_H$ time) and shall be less than the process determined statistical maximum limit as outlined in method 3131 of MIL-STD-750. See table III, subgroup 4 herein.

4.3.2 Power burn-in conditions. Power burn-in conditions are: $T_J$ = + 175 °C minimum, $V_{CB} \geq$ 100 V dc; $T_A$ = +30°C maximum.

4.3.3 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, “Discrete Semiconductor Die/Chip Lot Acceptance”. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.



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* 4.3.4 Dielectric withstanding voltage.

a. Magnitude of test voltage……..900 V dc.

b. Duration of application of test voltage……..15 seconds (min).

c. Points of application of test voltage……...All leads to case (bunch connection).

d. Method of connection……..Mechanical.

e. Kilovolt-ampere rating of high voltage source…….1,200 V/1.0 mA (min).

f. Maximum leakage current……...1.0 mA.

g. Voltage ramp up time.....500 V/second.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

* 4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 2037 | Test condition D, all internal leads for each device shall be pulled separately. |
| * B4 | 1037 | $V_{CB} \geq 100$ V dc. |
| B5 | 1027 | $V_{CB}$ = 100 V dc; adjust $T_A$ and $P_D$ to achieve $T_J$ = +275°C minimum. $P_D$ = 100 percent of rated $P_T$ minimum; $T_A$ = +100°C maximum. |

4.4.2.2 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1027 | For eutectic die attach: $V_{CB} \geq 100$ V dc; adjust $P_T$ to achieve $T_J$ = +175°C minimum; $T_A$ = +30°C maximum |
| B3 | 1037 | For solder die attach: 2,000 cycles, $V_{CB} \geq 100$ V dc. |



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* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

| | Subgroup | Method | Condition |
|---|---|---|---|
| | C2 | 2036 | Test condition A; weight = 10 pounds; time = 15 s. |
| | C2 | 2036 | Test condition D1; torque = 6 inch-ounce; time =15 s (2N6691, 2N6693 only). |
| | C2 | 2036 | Stud torque (2N6691, 2N6693 only), test condition D2; torque = 15 inch-pound; time = 15 s. |
| | C5 | 3131 | See 4.3.1, $R_{\theta JC}$ = 1.0°C for 2N6676, 2N6676T1, 2N6678, 2N6678T1, 2N6691, 2N6693; $R_{\theta JC}$ = 1.3°C for 2N6676T3, 2N6678T3. |
| | C6 | 1026 | For eutectic die attach: $V_{CB} \geq 100$ V dc; adjust $P_T$ to achieve $T_J$ = +175°C minimum; $T_A$ = +30°C maximum. |
| * | C6 | 1037 | For solder die attach: $V_{CB} \geq 100$ V dc. |

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS, JANJ, and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2 herein), except group D, subgroup 2 may be performed separate from other subgroups.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Insulation resistance test. Insulation resistance test conditions are as follows: Method 1016 of MIL-STD-750, short collector, emitter and base terminals together. Limit is $10^9$ Ω minimum.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3131 | See 4.3.1 | $Z_{\theta JX}$ | | | °C/W |
| Collector to base breakdown voltage | 3011 | Bias condition D, $I_C$ = 200 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N6676, 2N6676T1 2N6676T3, 2N6691 | | | | 300 | | V dc |
| 2N6678, 2N6678T1 2N6678T3, 2N6693 | | | | 400 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX1}$ | | 1 | µA dc |
| 2N6676, 2N6676T1 2N6676T3, 2N6691 | | $V_{CEX}$ = 450 V dc | | | | |
| 2N6678, 2N6678T1 2N6678T3, 2N6693 | | $V_{CEX}$ = 650 V dc | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX2}$ | | | |
| 2N6676, 2N6676T1 2N6676T3, 2N6691 | | $V_{CEX}$ = 300 V dc | | | 500 | nA |
| 2N6678, 2N6678T1 2N6678T3, 2N6693 | | $V_{CEX}$ = 400 V dc | | | 500 | nA |
| Collector to base cutoff current | 3036 | Bias condition D; | $I_{CBO}$ | | 1.0 | mA dc |
| 2N6676, 2N6676T1 2N6676T3, 2N6691 | | $V_{CBO}$ = 450 V dc | | | | |
| 2N6678, 2N6678T1 2N6678T3, 2N6693 | | $V_{CBO}$ = 650 V dc | | | | |
| Emitter-base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 8 V dc | $I_{EBO}$ | | 2.0 | mA dc |
| Base emitter voltage | 3066 | Test condition A; $I_C$ = 15 A dc; pulsed (see 4.5.1); $I_B$ = 3 A dc | $V_{BE(sat)}$ | | 1.5 | V dc |
| Collector to emitter saturated voltage | 3071 | $I_C$ = 15 A dc; pulsed (see 4.5.1) $I_B$ = 3 A dc | $V_{CE(sat)1}$ | | 1.0 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 1 A dc; pulsed (see 4.5.1) | $h_{FE1}$ | 15 | 40 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 15 A dc; pulsed (see 4.5.1) | $h_{FE2}$ | 8 | 20 | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_A$ = +125°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A; $V_{BE}$ = -1.5 V dc | $I_{CEX3}$ | | 50 | µA dc |
| 2N6676, 2N6676T1<br>2N6676T3, 2N6691 | | $V_{CEX}$ = 450 V dc | | | | |
| 2N6678, 2N6678T1<br>2N6678T3, 2N6693 | | $V_{CEX}$ = 650 V dc | | | | |
| Collector to emitter saturated voltage | 3071 | $I_C$ = 15 A dc; $I_B$ = 3 A dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 2.0 | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX4}$ | | | |
| 2N6676, 2N6676T1<br>2N6676T3, 2N6691 | | $V_{CEX}$ = 300 V dc | | | 90 | µA |
| 2N6678, 2N6678T1<br>2N6678T3, 2N6693 | | $V_{CEX}$ = 400 V dc | | | 90 | µA |
| Low-temperature operation : | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc $I_C$ = 15 A dc; pulsed (see 4.5.1) | $h_{FE3}$ | 4 | | |
| Subgroup 4 | | | | | | |
| Magnitude of common emitter small-signal short-circuit forward- current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = 1 A dc; f = 5 MHz | $\|h_{fe}\|$ | 3 | 10 | |
| Open capacitance (open circuit) | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1.0 MHz | $C_{obo}$ | 150 | 500 | pF |
| Pulse response: | 3251 | $T_A$ = +25°C; Test condition A, except test circuit and pulse requirement in accordance with figure 12 | | | | |
| Pulse delay time | | | $t_d$ | | 0.1 | µs |
| Pulse rise time | | | $t_r$ | | 0.6 | µs |
| Pulse storage time | | | $t_s$ | | 2.5 | µs |
| Pulse fall time | | | $t_f$ | | 0.5 | µs |
| Cross over time | | | $t_c$ | | 0.5 | µs |
| Subgroup 5 | | $T_C$ = +25°C | | | | |
| Safe operating area (dc operation) | 3051 | t = 1 s; 1 cycle; (see figure 13) | | | | |
| Test 1<br>(All device types) | | $V_{CE}$ = 11.7 V dc; $I_C$ = 15 A dc | | | | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 - Continued | | | | | | |
| Test 2<br>(2N6676, 2N6678, 2N6676T1, 2N6676T3, 2N6678T1, 2N6678T3) | | $V_{CE}$ = 30 V dc; $I_C$ = 5.9 A dc | | | | |
| Test 3<br>(All device types) | | $V_{CE}$ = 100 V dc; $I_C$ = 0.25 A dc | | | | |
| Test 4<br>(2N6691, 2N6693) | | $V_{CE}$ = 25 V dc; $I_C$ = 7 A dc | | | | |
| Test 5<br>(2N6676, 2N6678, 2N6676T1, 2N6676T3, 2N6678T1, 2N6678T3 )<br>(2N6691, 2N6693) | | $V_{CE}$ = 300 V dc; $I_C$ = 20 mA dc<br>$V_{CE}$ = 400 V dc; $I_C$ = 10 mA dc | | | | |
| Safe operating area (clamped switching) | 3053 | $T_A$ = +25°C, $V_{CC}$ = 15 V dc (see figure 14 ); Load condition B, $V_{BB2}$ = 5.0 V, $R_{BB1}$ = 5 Ω; $R_{BB2}$ = 1.5 Ω; L = 50 μH; R of inductor = 0.05 Ω, $R_{load}$ =R of inductor | | | | |
| 2N6676, 2N6676T1, 2N6676T3, 2N6691 | | Clamp voltage = 350 V dc; $I_C$ = 15 A dc | | | | |
| 2N6678, 2N6678T1, 2N6678T3, 2N6693 | | Clamp voltage = 450 V dc; $I_C$ = 15 A dc | | | | |
| Electrical measurements | | Table I, subgroup 2 herein. | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Insulation resistance ( 2N6676T1, 2N6676T3, 2N6678T1, 2N6678T3, 2N6691 and 2N6693 only) | 1016 | See 4.5.2 | $R_{ISO}$ | $1x10^9$ | | Ω |

1/ For sampling plan see MIL-PRF-19500.



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TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 4/ | | | | | | |
| Neutron irradiation | 1017 | Neutron exposure Vces = 0V | | | | |
| Collector to base breakdown voltage | 3011 | Bias condition D, $I_C$ = 200 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N6676, 2N6676T1 2N6676T3, 2N6691 | | | | 300 | | V dc |
| 2N6678, 2N6678T1 2N6678T3, 2N6693 | | | | 400 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX1}$ | | 2 | µA dc |
| 2N6676, 2N6676T1 2N6676T3, 2N6691 | | $V_{CEX}$ = 450 V dc | | | | |
| 2N6678, 2N6678T1 2N6678T3, 2N6693 | | $V_{CEX}$ = 650 V dc | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX2}$ | | | |
| 2N6676, 2N6676T1 2N6676T3, 2N6691 | | $V_{CEX}$ = 300 V dc | | | 1.0 | uA |
| 2N6678, 2N6678T1 2N6678T3, 2N6693 | | $V_{CEX}$ = 400 V dc | | | 1.0 | uA |
| Collector to base cutoff current | 3036 | Bias condition D; | $I_{CBO}$ | | 2.0 | mA dc |
| 2N6676, 2N6676T1 2N6676T3, 2N6691 | | $V_{CBO}$ = 450 V dc | | | | |
| 2N6678, 2N6678T1 2N6678T3, 2N6693 | | $V_{CBO}$ = 650 V dc | | | | |
| Emitter-base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 8 V dc | $I_{EBO}$ | | 4.0 | mA dc |
| Base emitter voltage | 3066 | Test condition A; $I_C$ = 15 A dc; pulsed (see 4.5.1); $I_B$ = 3 A dc | $V_{BE(sat)}$ | | 1.73 | V dc |
| Collector to emitter saturated voltage | 3071 | $I_C$ = 15 A dc; pulsed (see 4.5.1) $I_B$ = 3 A dc | $V_{CE(sat)1}$ | | 1.15 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 1 A dc; pulsed (see 4.5.1) | [$h_{FE1}$] 5/ | [7.5] | 40 | |
| Forward-current transfer Ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 15 A dc; pulsed (see 4.5.1) | [$h_{FE2}$] 5/ | [4] | 20 | |

See footnotes at end of table.



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TABLE II. Group D inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 | | | | | | |
| Total dose irradiation | 1019 | Gamma exposure | | | | |
| 2N6676, 2N6676T1<br>2N6676T3, 2N6691 | | Vces = 240V | | | | |
| 2N6678, 2N6678T1<br>2N6678T3, 2N6693 | | Vces = 320V | | | | |
| Collector to base breakdown voltage | 3011 | Bias condition D, $I_C$ = 200 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N6676, 2N6676T1<br>2N6676T3, 2N6691 | | | | 300 | | V dc |
| 2N6678, 2N6678T1<br>2N6678T3, 2N6693 | | | | 400 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX1}$ | | 2 | µA dc |
| 2N6676, 2N6676T1<br>2N6676T3, 2N6691 | | $V_{CEX}$ = 450 V dc | | | | |
| 2N6678, 2N6678T1<br>2N6678T3, 2N6693 | | $V_{CEX}$ = 650 V dc | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX2}$ | | | |
| 2N6676, 2N6676T1<br>2N6676T3, 2N6691 | | $V_{CEX}$ = 300 V dc | | | 1.0 | uA |
| 2N6678, 2N6678T1<br>2N6678T3, 2N6693 | | $V_{CEX}$ = 400 V dc | | | 1.0 | uA |
| Collector to base cutoff current | 3036 | Bias condition D; | $I_{CBO}$ | | 2.0 | mA dc |
| 2N6676, 2N6676T1<br>2N6676T3, 2N6691 | | $V_{CBO}$ = 450 V dc | | | | |
| 2N6678, 2N6678T1<br>2N6678T3, 2N6693 | | $V_{CBO}$ = 650 V dc | | | | |
| Emitter-base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 8 V dc | $I_{EBO}$ | | 4.0 | mA dc |
| Base emitter voltage | 3066 | Test condition A; $I_C$ = 15 A dc; pulsed (see 4.5.1); $I_B$ = 3 A dc | $V_{BE(sat)}$ | | 1.73 | V dc |
| Collector to emitter saturated voltage | 3071 | $I_C$ = 15 A dc; pulsed (see 4.5.1)<br>$I_B$ = 3 A dc | $V_{CE(sat)1}$ | | 1.15 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 1 A dc; pulsed (see 4.5.1) | $[h_{FE1}]$ 5/ | [7.5] | 40 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 3 V dc; $I_C$ = 15 A dc; pulsed (see 4.5.1) | $[h_{FE2}]$ 5/ | [4] | 20 | |

See footnotes at end of table.



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TABLE II. Group D inspection - Continued.

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to all device types unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019, of MIL-STD-750, for how to determine [$h_{FE}$] by first calculating the delta($1/h_{FE}$) from the pre and post radiation $h_{FE}$. Notice that [$h_{FE}$] is not the same as $h_{FE}$ and cannot be measured directly. The [$h_{FE}$] value can never exceed the pre-radiation minimum $h_{FE}$ that it is based upon.



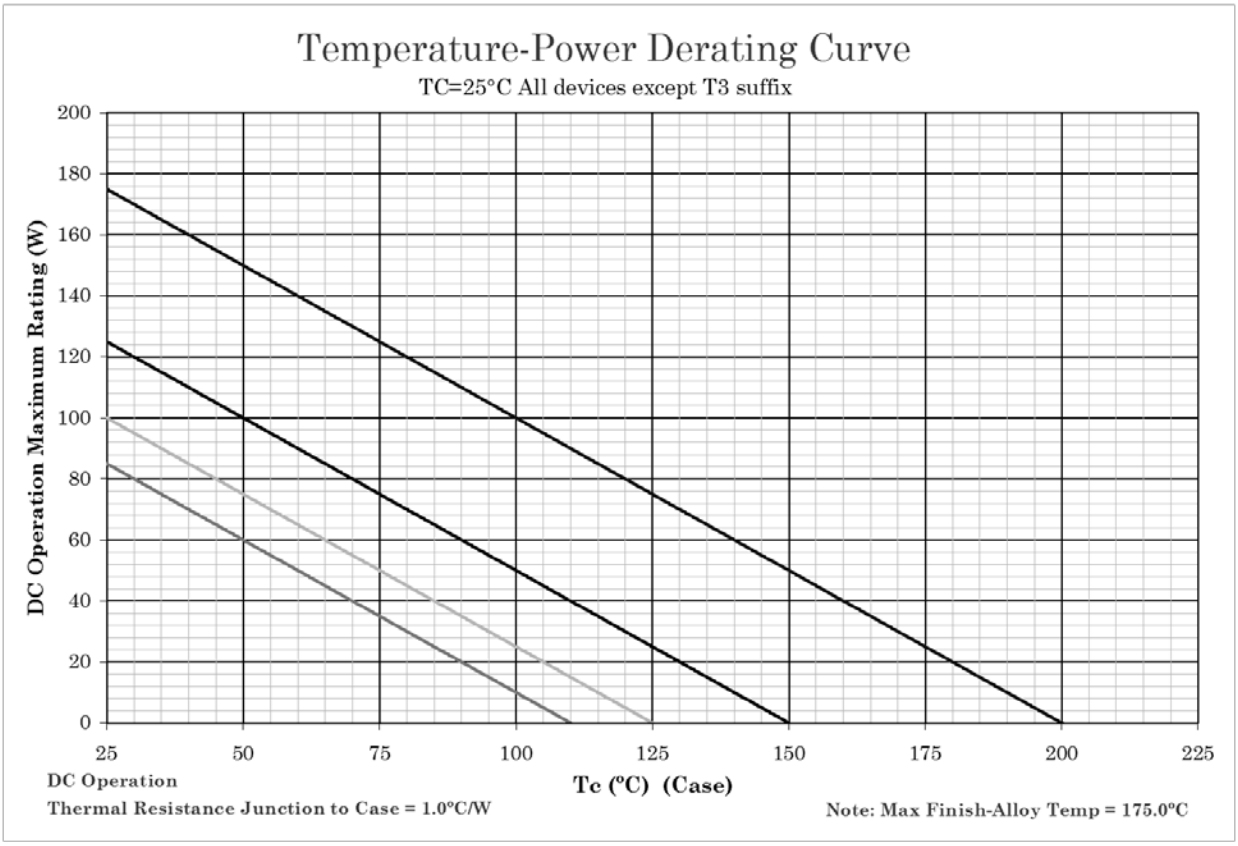
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* TABLE III. Group E inspection (all quality levels) - for qualification and re-qualification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Thermal shock<br>glass strain | 1056 | 0°C to +100°C, 100 cycles. | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Blocking life | 1048 | 1,000 hours minimum, $T_A$ = 150°C, $V_{CB}$ = 80 percent of rated, without going over the maximum rated $V_{CEO}$. | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 4 | | | |
| Thermal impedance curves | | See table E-IX of MIL-PRF-19500. | |
| Subgroup 5 | | | 3 devices<br>c = 0 |
| Barometric pressure | 1001 | Test condition C, see 1.3. Unless otherwise specified, the device shall be subjected to the maximum voltage it would be subjected to under rated operating conditions. | |
| Subgroup 6 | | | |
| * ESD | 1020 | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition A. | |



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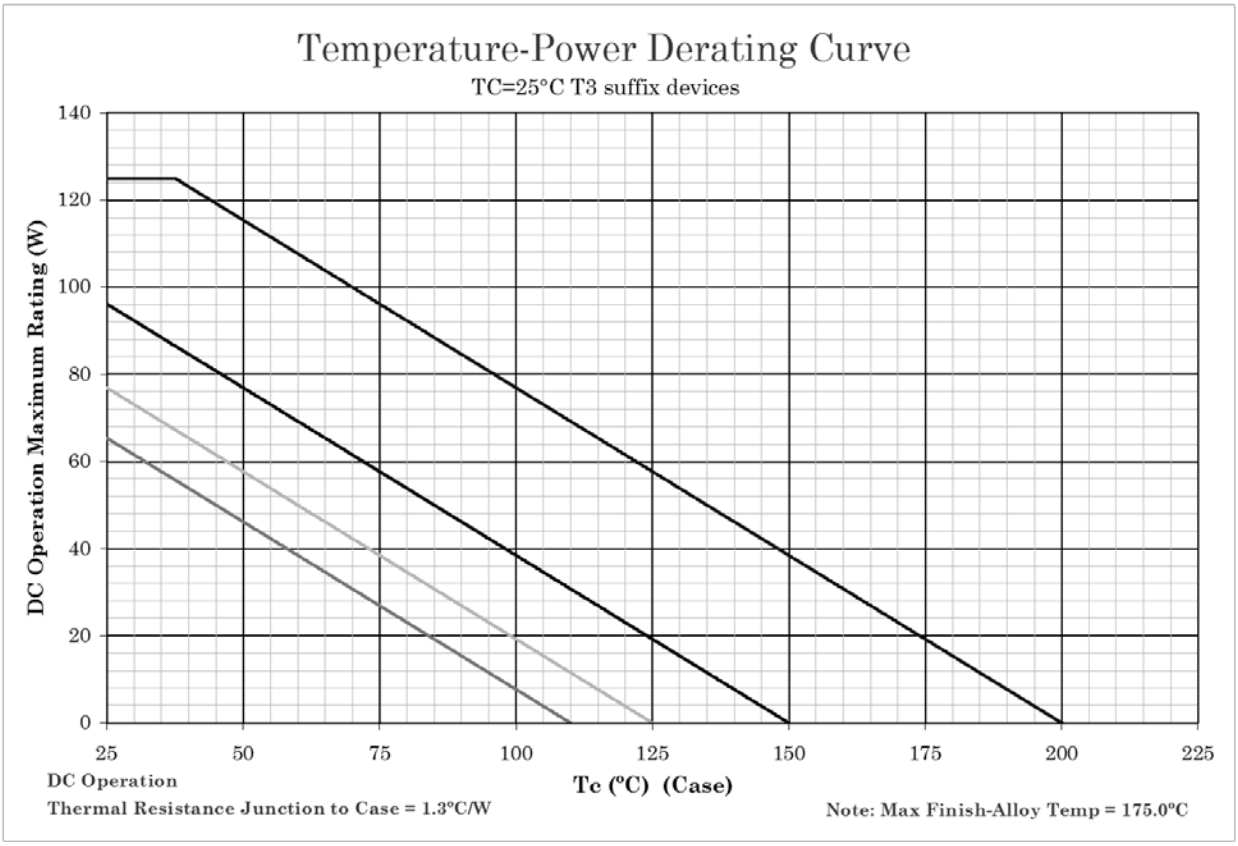

NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 6. Temperature derating graph (all except T3 (TO-257AA) packages).



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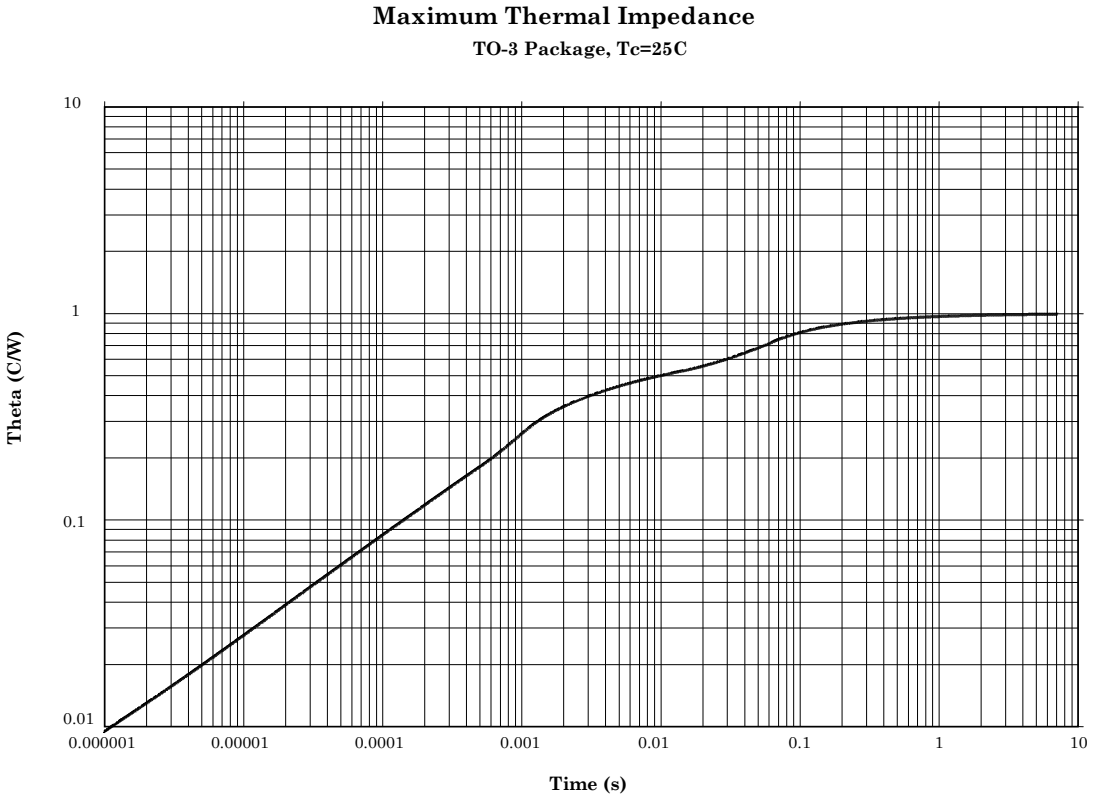

NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 7. Temperature derating graph for T3 (TO-257AA) packages.



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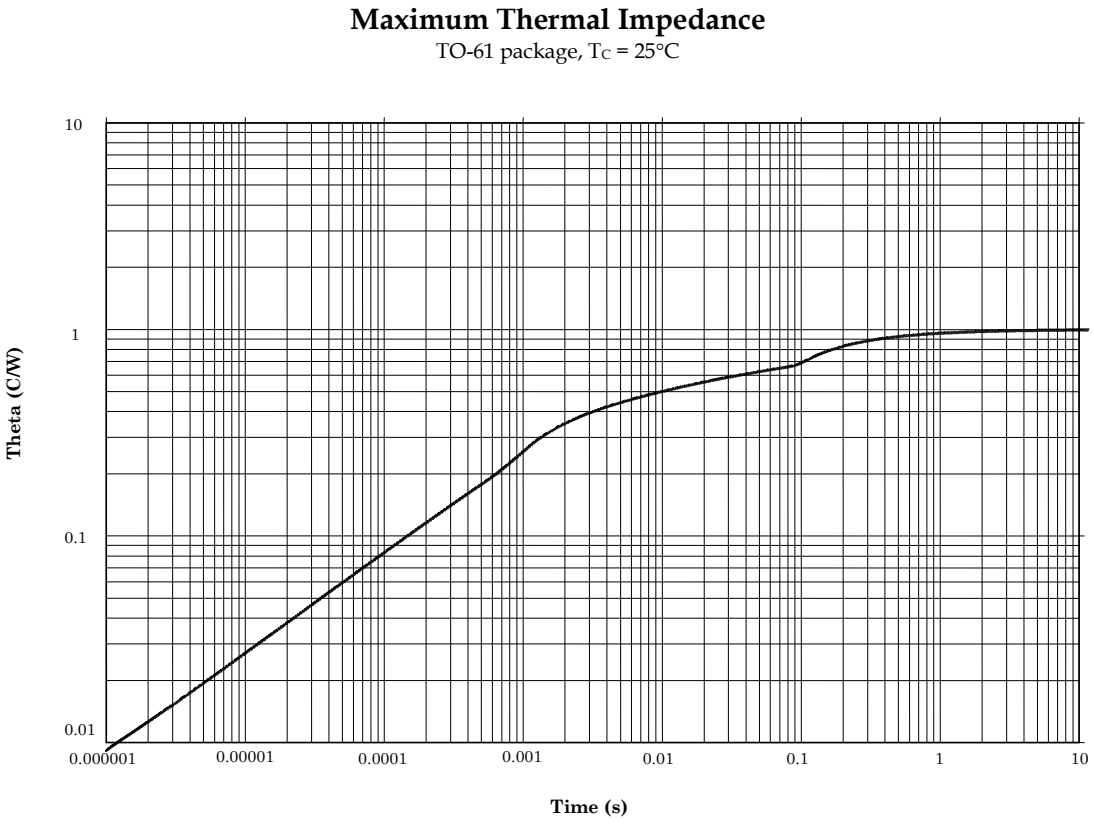



$R_{\theta JC}$ = 1.0 °C/W max.

FIGURE 8. Thermal impedance graphs (2N6676 and 2N6678).



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$R_{\theta JC}$ = 1.0 °C/W max.

FIGURE 9. Thermal impedance graphs (2N6691 and 2N6693).



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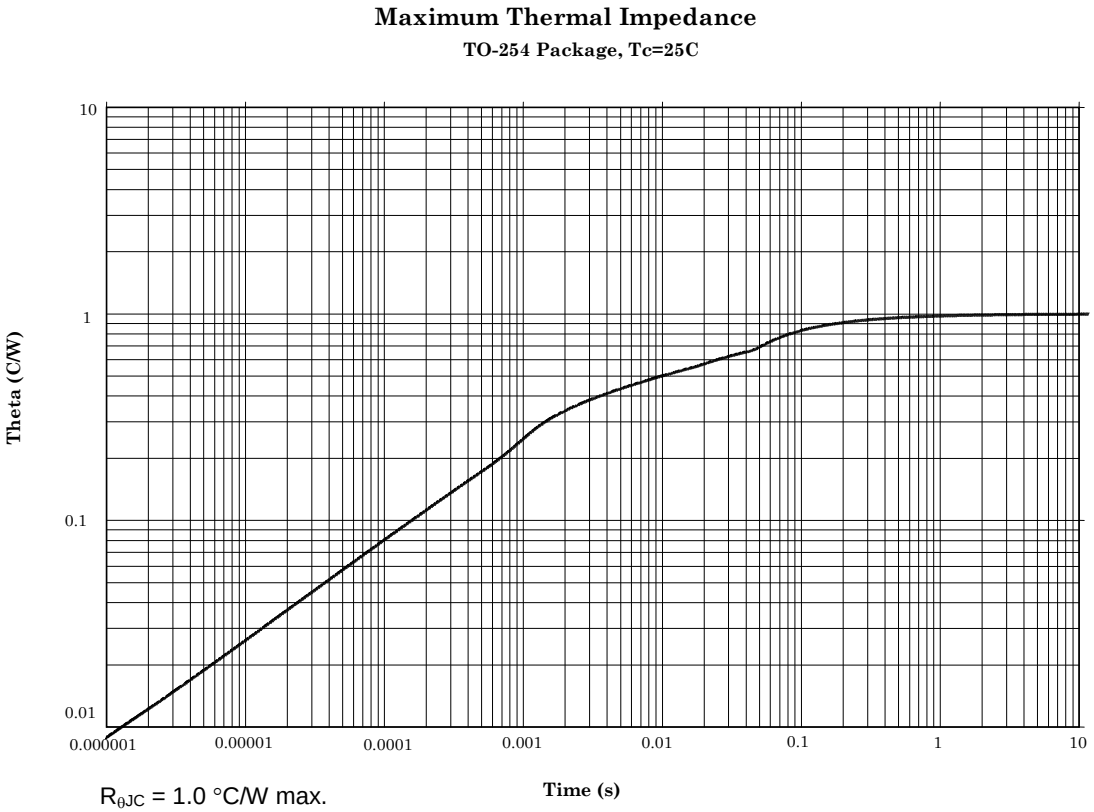
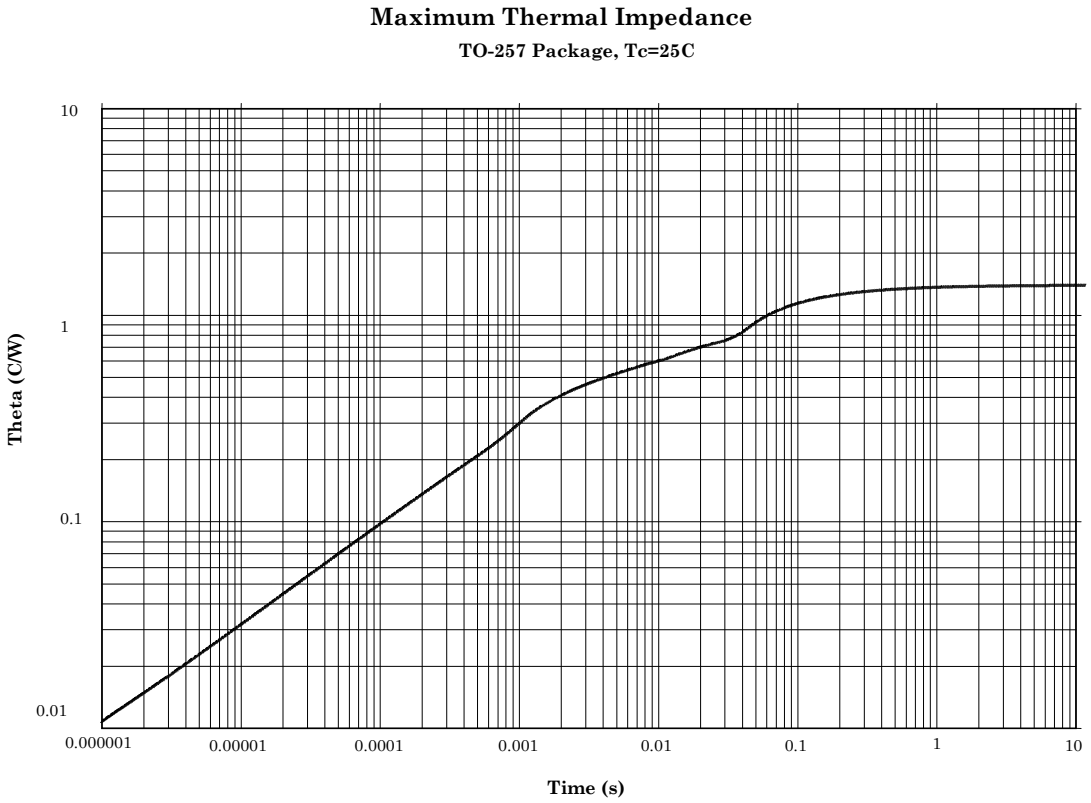



FIGURE 10. Thermal impedance graphs (2N6676T1 and 2N6678T1).



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$R_{\theta JC}$ = 1.3 °C/W max.

FIGURE 11. Thermal impedance graphs (2N6676T3 and 2N6678T3).



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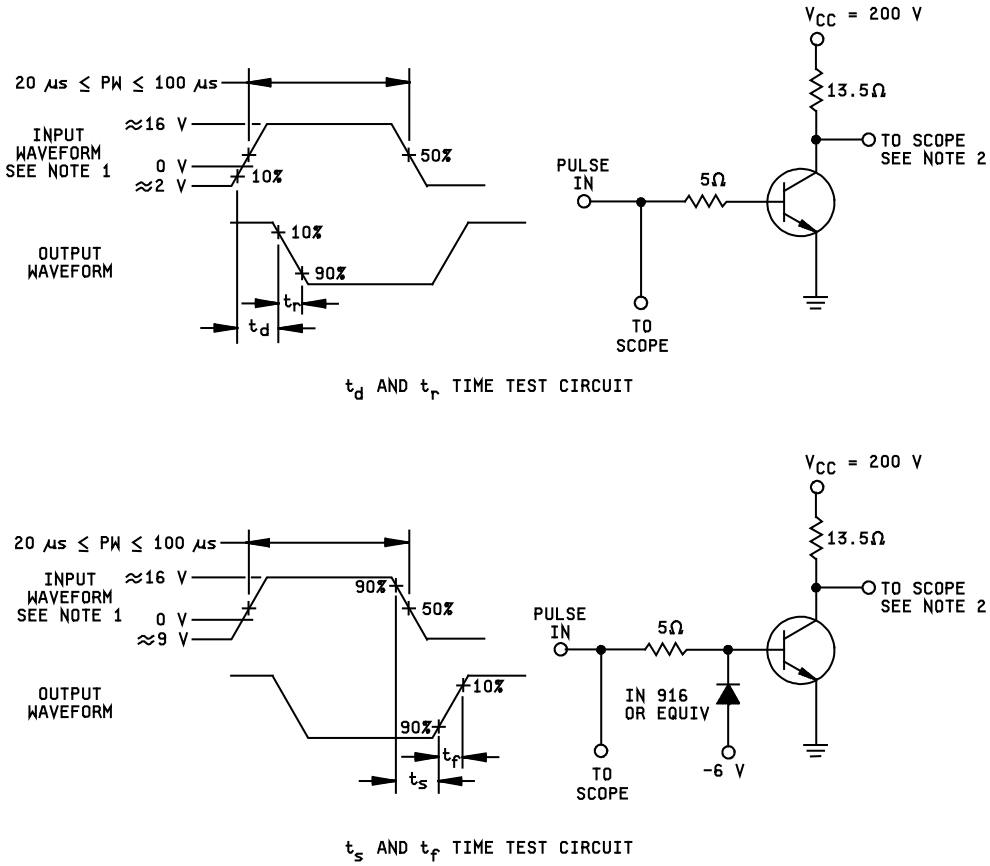
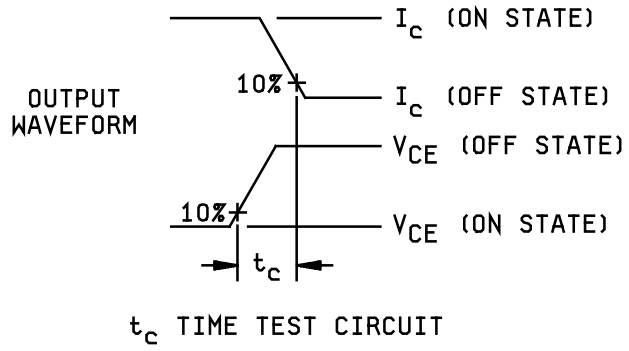


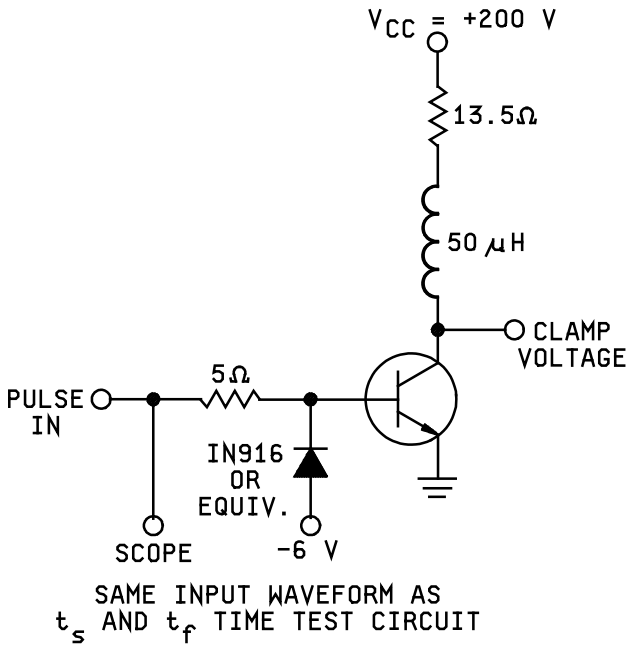

FIGURE 12. Pulse response test circuit.



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$t_c$ TIME TEST CIRCUIT




SAME INPUT WAVEFORM AS $t_s$ AND $t_f$ TIME TEST CIRCUIT

NOTES:

1. The rise time ($t_r$ ) of the applied pulse shall be ≤ 20 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 ohms.
2. Output sampling oscilloscope: $Z_{in} \geq 100$ k ohms; $C_{in} \leq 12$ pF; rise time ≤ 5 ns.

FIGURE 12. Pulse response test circuits - Continued.



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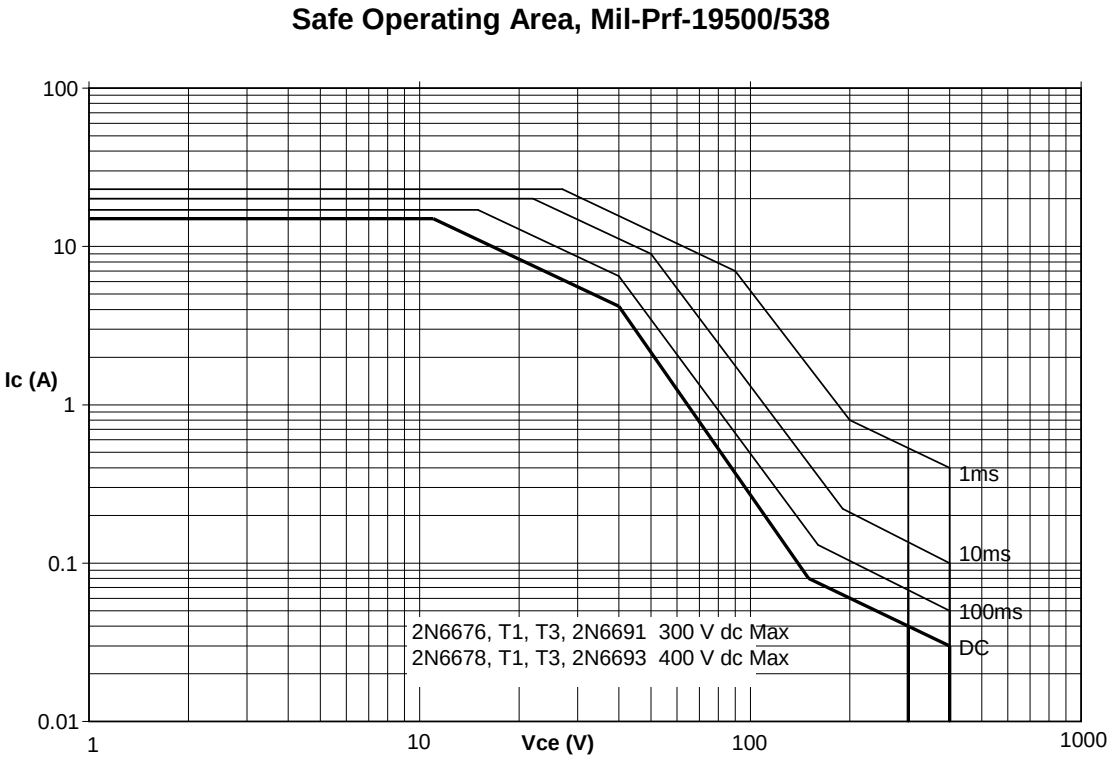



FIGURE 13. Maximum safe operating graph (dc).



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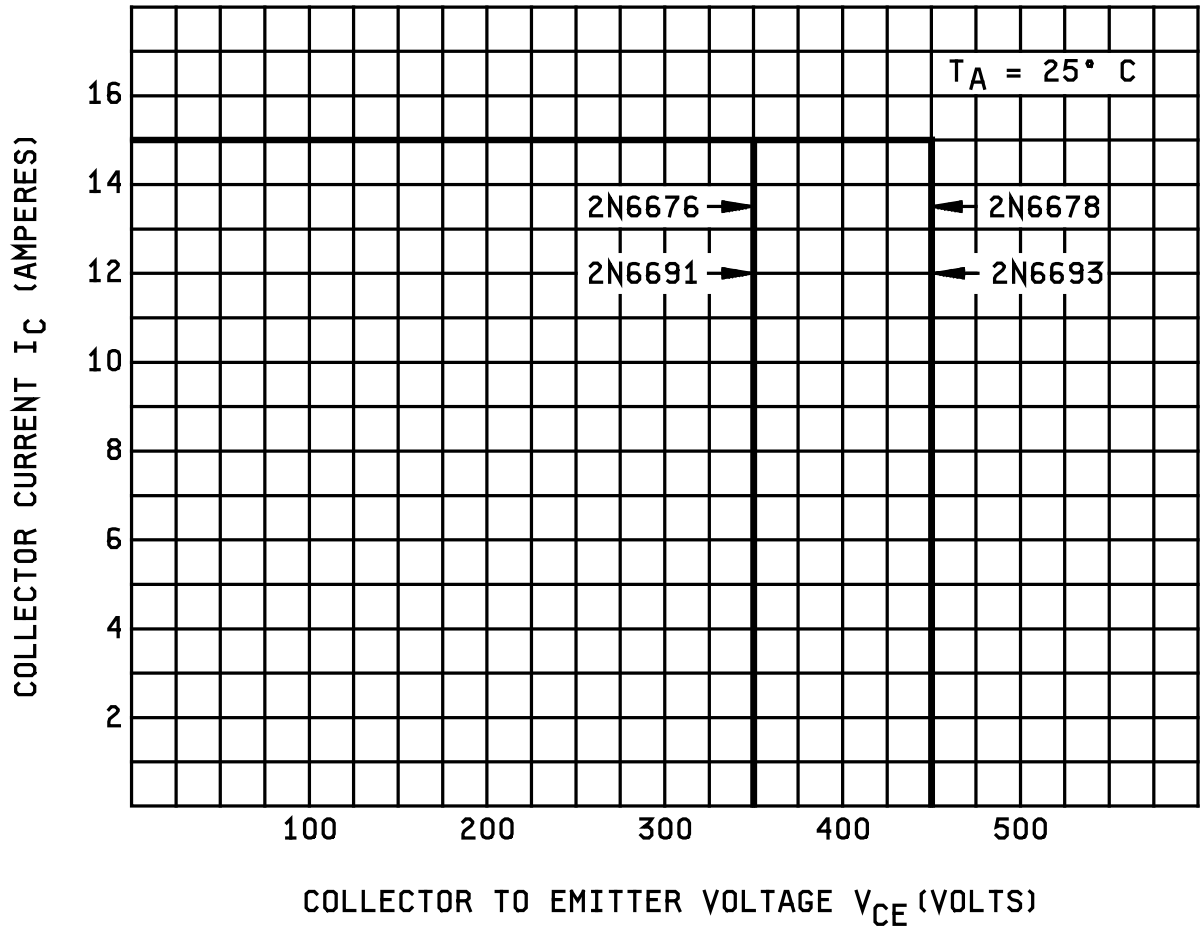



FIGURE 14. Safe operating area for switching between saturation and cutoff (clamped inductive load) (all devices).



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it should be specified in the contract.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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6.5 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHC2N2434A) will be identified on the QML.

| JANC ordering information | |
|---|---|
| PIN | Manufacturer |
| | 43611 |
| 2N6676 | JANHCA2N6676<br>JANKCA2N6676 |
| 2N6678 | JANHCA2N6678<br>JANKCA2N6678 |

Custodians:
Army - CR
Air Force - 85
DLA - CC

Review activities:
Army - MI
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2014-029)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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